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GitHub Repository: https://github.com/Thotavasu/svm-kernel-tutorial

Understanding Support Vector Machines: A Comparative Study of Kernel Functions and Their Impact on Classification Performance

1. Introduction

Support Vector Machines (SVMs) are widely used supervised machine learning models designed for classification tasks. Their primary goal is to identify an optimal decision boundary that separates data points belonging to different classes. Unlike simpler models, SVMs aim to maximise the margin between classes, which improves generalisation and robustness.

A key strength of SVMs is their ability to handle both linear and non-linear data through the use of kernel functions. These kernels enable the model to transform data into higher-dimensional spaces, allowing complex patterns to be captured without explicitly increasing computational complexity.

This tutorial focuses on analysing how different kernel functions—specifically linear, polynomial, and radial basis function (RBF) kernels—affect classification performance. Experiments are conducted using both a real-world dataset and a synthetic dataset to highlight differences in behaviour.

The aim is to provide a clear understanding of when and why different kernels should be used in practice. This tutorial aims not only to demonstrate the implementation of SVMs, but also to provide practical insight into how different kernels influence model behaviour and performance in real-world applications.

2. Support Vector Machines

Support Vector Machines (SVMs) are supervised learning models used for classification by constructing an optimal decision boundary between data points belonging to different classes. The main objective of an SVM is to find a boundary that not only separates the data but does so in a way that maximises the margin between classes, leading to better generalisation on unseen data.

2.1 Hyperplane and Margin

In SVM, the decision boundary is referred to as a hyperplane. In a two-dimensional space, this is simply a straight line, while in higher dimensions it becomes a plane or a hyper-surface. The position of this hyperplane is determined such that it separates the classes as effectively as possible.

A key concept in SVM is the *margin*, which is defined as the distance between the hyperplane and the nearest data points from each class. Instead of simply finding any separating boundary, SVM aims to maximise this margin. A larger margin reduces the model's sensitivity to small variations in the data and improves its ability to generalise to new, unseen samples.

2.2 Support Vectors

Not all data points influence the position of the decision boundary. Only a subset of points, known as *support vectors*, play a critical role in defining the hyperplane. These are the data points that lie closest to the decision boundary.

Because the model depends only on these critical points, SVMs are memory efficient and robust

to noise in the dataset. However, this also means that incorrect or noisy data near the boundary can significantly affect the model's performance.

3. Kernel Trick

While Support Vector Machines perform well on linearly separable data, real-world datasets are often more complex and cannot be separated by a straight line or simple hyperplane. To address this limitation, SVMs use a technique known as the *kernel trick*, which enables them to handle non-linear relationships effectively.

The kernel trick works by implicitly mapping the original input data into a higher- dimensional feature space, where a linear separation becomes possible. Importantly, this transformation is performed without explicitly computing the coordinates in the higher- dimensional space, which makes the process computationally efficient. Different kernel functions define how this transformation is carried out, and the choice of kernel has a significant impact on model performance.

3.1 Linear Kernel

The linear kernel is the simplest form of kernel function and does not transform the data into a higher-dimensional space. Instead, it attempts to find a straight-line (or hyperplane) decision boundary in the original feature space.

This kernel is effective when the data is approximately linearly separable. However, it struggles when the relationship between features and classes is more complex, leading to underfitting.

3.2 Polynomial Kernel

The polynomial kernel introduces non-linearity by considering polynomial combinations of the input features. This allows the model to create more flexible decision boundaries compared to the linear kernel.

The degree of the polynomial controls the complexity of the boundary. While higher degrees can capture more intricate patterns, they also increase the risk of overfitting, especially when the dataset is small or noisy.

3.3 Radial Basis Function (RBF) Kernel

The Radial Basis Function (RBF) kernel is one of the most widely used kernels due to its flexibility and strong performance across a wide range of problems. It maps data into an infinite-dimensional space, allowing the model to form highly complex decision boundaries.

The behaviour of the RBF kernel is controlled by a parameter called *gamma*, which determines how far the influence of a single training point extends. A small gamma results in smoother decision boundaries, while a large gamma can lead to highly complex boundaries that may overfit the data.

This demonstrates that the kernel trick is essential for extending the capability of SVMs beyond simple linear problems, enabling them to model complex real-world data effectively.

3.4 Importance of Kernel Selection

Choosing the appropriate kernel is critical for achieving good performance. A kernel that is too simple may fail to capture the underlying structure of the data, while an overly complex kernel may fit noise rather than meaningful patterns.

In this tutorial, different kernels are evaluated using both real-world and synthetic datasets to demonstrate how their behaviour changes depending on the data structure. This comparison provides insight into selecting the most suitable kernel for a given problem.

4. Datasets

To evaluate the performance and behaviour of Support Vector Machines, both a real-world dataset and a synthetic dataset were used. This combination allows for both practical evaluation and clear visual interpretation of model behaviour.

4.1 Heart Disease Dataset

The primary dataset used in this study is a heart disease dataset, which is commonly used for binary classification tasks. The objective is to predict whether a patient is likely to have heart disease based on a set of medical attributes.

The dataset contains multiple numerical features representing clinical measurements such as age, blood pressure, cholesterol levels, and other health indicators. The target variable represents the presence or absence of heart disease.

This dataset provides a realistic scenario where SVM models can be evaluated based on their predictive performance and generalisation ability.

4.2 Synthetic Moons Dataset

In addition to the real-world dataset, a synthetic dataset known as the “moons dataset” was used. This dataset consists of two interleaving half-circle shapes, making it inherently non-linearly separable.

The purpose of using this dataset is to visually demonstrate how different kernel functions affect the shape of the decision boundary. Unlike the heart dataset, the moons dataset allows direct visualisation of how SVM models classify data.

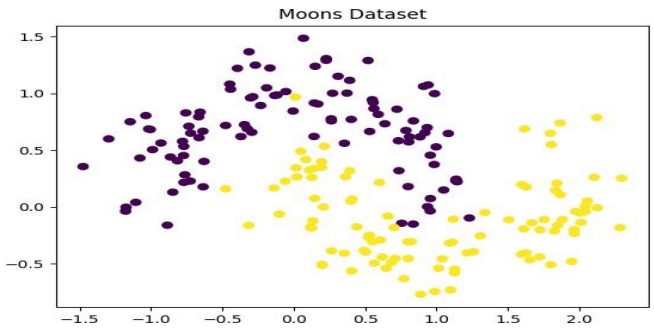

Figure 1: Distribution of the synthetic moons dataset showing non-linear class separation.

5. Experimental Setup

To ensure a fair evaluation of the models, the dataset was divided into training and testing subsets using an 80:20 split. This allows the performance of the models to be evaluated on unseen data.

Feature scaling was applied using standardisation to ensure that all input features contribute equally to the model. This step is particularly important for SVMs, as they rely on distance-based calculations.

Three different kernel functions were evaluated:

- Linear kernel
- Polynomial kernel
- Radial Basis Function (RBF) kernel

Additionally, hyperparameter tuning was performed using grid search to identify the optimal values of key parameters such as C and gamma.

6. Results and Analysis

6.1 Kernel Comparison

The performance of the SVM models varied depending on the choice of kernel. The linear kernel provided a baseline performance but struggled to capture complex relationships in the data.

The polynomial kernel improved performance by introducing non-linearity, but its effectiveness depended on the chosen degree. The RBF kernel consistently provided the best results, as it is capable of modelling highly complex decision boundaries.

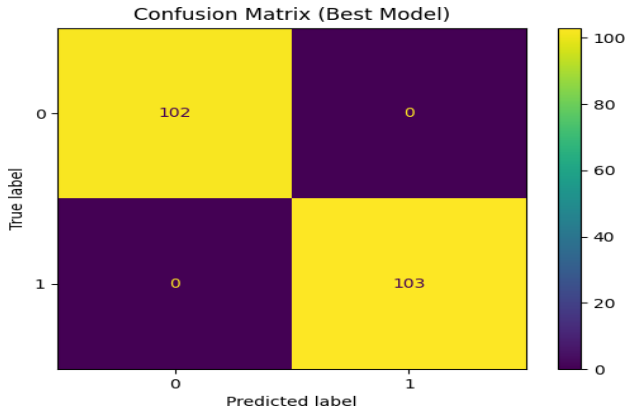

Figure 2: Confusion matrix showing classification performance of the optimised SVM model.

The results demonstrate that kernel selection plays a critical role in determining the effectiveness of Support Vector Machines. The linear kernel provides a simple decision boundary, but its performance is limited when the underlying data is not linearly separable. The polynomial kernel introduces additional flexibility, allowing the model to capture more complex relationships, although its effectiveness depends on the chosen degree.

In contrast, the RBF kernel achieves superior performance by adapting to the structure of the data and forming highly flexible decision boundaries. This demonstrates that the success of SVM models is not solely dependent on the algorithm itself, but on how well the chosen kernel aligns with the complexity of the dataset.

6.2 Parameter Tuning

Hyperparameter tuning using grid search significantly improved model performance. The parameter C was found to control the trade-off between margin size and classification accuracy, while gamma influenced the flexibility of the decision boundary.

Higher values of gamma resulted in more complex decision boundaries, which could lead to overfitting, whereas lower values produced smoother boundaries with better generalisation.

This highlights the importance of carefully selecting both the kernel function and its parameters when applying SVMs in practical scenarios.

7. Visual Demonstration

To better understand the effect of different kernels, the decision boundaries shown in Figures 3–5 illustrate how each kernel models the data.

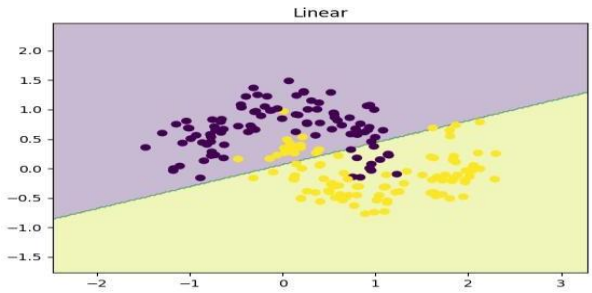

Figure 3: Decision boundary using linear kernel (underfitting observed)

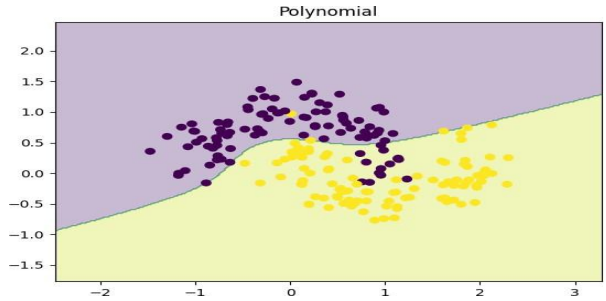

Figure 4: Decision boundary using polynomial kernel

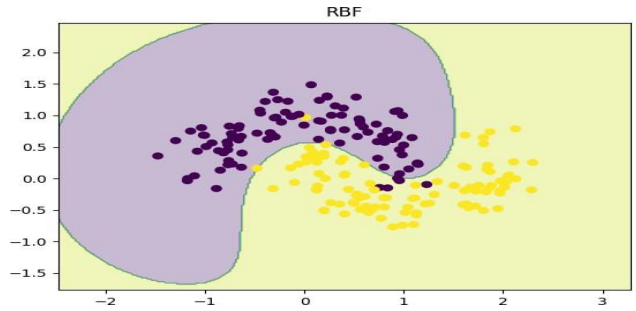

Figure 5: Decision boundary using RBF kernel (best fit)

The visualisations provide a clear comparison of how different kernel functions influence the shape of the decision boundary. The linear kernel produces a straight boundary, which fails to separate the data effectively due to the non-linear structure of the dataset. This results in underfitting, where the model is unable to capture important patterns. The polynomial kernel introduces curvature into the boundary, improving the model's ability to represent more complex relationships. However, it still does not fully align with the data distribution, indicating that its flexibility is limited by the chosen degree.

The RBF kernel produces a smooth and highly adaptive decision boundary that closely follows the data points. This results in more accurate classification, as the model successfully captures the non-linear structure of the dataset. These observations demonstrate that kernel choice directly affects the model's ability to learn complex patterns, making it a crucial factor in SVM performance. This demonstrates that visual analysis is a powerful tool for understanding model behaviour and validating performance.

8. Critical Discussion

Support Vector Machines offer several advantages, including strong performance in high-dimensional spaces and robustness to overfitting when properly tuned. Their reliance on support vectors makes them efficient in terms of memory usage and effective for complex classification tasks.

However, SVMs also have notable limitations. They are highly sensitive to parameter selection and require careful tuning of hyperparameters such as C and gamma. Additionally, SVMs do not scale well to very large datasets due to their computational complexity.

Another limitation is the difficulty in selecting an appropriate kernel function, as an incorrect choice can lead to poor generalisation. This highlights the importance of understanding both the data and the model when applying SVMs in real-world scenarios.

9. Ethical Considerations

The use of machine learning in healthcare applications raises important ethical concerns. In the context of heart disease prediction, incorrect classifications could lead to serious consequences, including misdiagnosis or delayed treatment.

Bias in the dataset may also result in unfair predictions for certain groups. Therefore, it is essential to ensure that models are trained on diverse and representative data, and that their predictions are interpreted with caution.

9.1 Accessibility Considerations

To ensure accessibility, this tutorial incorporates several design considerations. All visualisations use clear colour contrasts and labelled axes to improve readability, including for individuals with colour vision deficiencies. Each figure is accompanied by a descriptive caption to ensure that its purpose is understandable without relying solely on visual interpretation.

The document structure follows a clear and consistent format, making it easier to navigate and compatible with assistive technologies such as screen readers. These considerations ensure that the tutorial can be effectively understood by a wide range of users.

10. Conclusion

This tutorial demonstrated how Support Vector Machines can be used for classification tasks and how different kernel functions influence model performance.

Through both real-world and synthetic datasets, it was observed that the RBF kernel provides superior performance in handling non-linear data. The results highlight the importance of kernel selection and parameter tuning in achieving optimal performance.

Overall, the findings demonstrate that the effectiveness of Support Vector Machines is highly dependent on both kernel selection and parameter tuning. Selecting an appropriate kernel allows the model to better represent the underlying data structure, leading to improved classification performance. This is particularly important in real-world applications, such as healthcare, where accurate predictions are essential for informed decision-making.